Research Statement

Yanjin Li · yanjin2@illinois.edu · yjli-new.github.io

RESEARCH GOAL

My research goal is to build AI systems that run directly on local devices, such as personal computers, phones, and offline robots, and that keep improving through interaction with their environment. Today's frontier models rely on power-hungry cloud clusters and constant network connectivity, which limits their adoption in offline physical environments and on everyday personal devices. Reaching this goal requires progress on two fronts. First, algorithms and the systems beneath them must become efficient together, so that models fit the limited compute and energy budgets of edge devices. Second, capability must grow from sources beyond mining existing data: agents that improve themselves through real interaction, and model architectures beyond next-token prediction.

PAST WORK

Accelerating video diffusion without retraining. In *DecoupAttn*, a paper I completed independently, I **substantially accelerated vDiTs** by approximating the attention that sparse methods skip. When my first design gave no gain, I located the sources of error through ablations from several angles and revised the approach. The final method **outperforms 7 recent methods across 10 models**, achieving the best generation quality and inference speed.

Evolving agent skills with verified admission. I **wrote all of the core code and ran every evaluation** for R^2 *Flow*, which improves an agent by evolving its own skill library and reaches **state-of-the-art results on all 12 benchmarks**. I also co-developed *ScienceClaw*, an open-source agent framework with **more than 1,000 GitHub stars**, and built its replay-based admission check for new skills.

Building 3D perception and rendering systems. At Tsinghua University, I benchmarked light-transport algorithms on a CUDA path tracer, and in another group there I built an open-vocabulary 3D object detector that fuses camera and LiDAR input. These experiences gave me a lasting research habit: **first locate a model's computational bottlenecks and sources of error, then design algorithms and systems that target them.**

FUTURE RESEARCH

Allocating computation on demand across tokens and layers. Current models spend substantial redundant computation across attention, layers, and tokens. Extending the sparse-attention approximation in *DecoupAttn* from attention maps to whole layers and tokens, I aim to design models that decide for each input where exact computation is needed, fitting within the memory and power limits of mobile devices. Because skipped computation becomes real speedup only when kernels and memory layout work together, this direction calls for co-designing the model and its runtime.

Evolving the agent harness for lightweight online learning. Changing the weights of a pretrained model costs substantial GPU compute. By adjusting its harness, the tools, skills, and memory that surround a model, an agent can instead learn online in a lightweight way, with frozen weights and few or no gradient updates. Building on the skill evolution in R^2 *Flow* and the replay-based admission check in *ScienceClaw*, I aim to build self-evolving agents that judge from everyday interaction which changes to their tools and skills actually help, then verify and keep those changes automatically.

Reasoning directly over spatial and visual representations. Physical interaction rests on continuous spatial and visual signals, and it is not confined to discrete linguistic symbols. Drawing on my experience with CUDA path tracing and camera-and-LiDAR 3D detection, I plan to explore models whose internal representations are 3D scenes, rendered views, and graphs, so that reasoning happens directly in space and pixels. Alongside these models, I will design the low-level systems that run such representations efficiently on edge devices.

SUMMARY

Because these questions span machine learning, computer systems, and computer vision, I hope to collaborate with faculty across these areas. My mathematical training helps me bound errors and costs before I write code and read theory quickly, so I turn ideas into working systems faster. Over the next five to ten years, my goal is to bring these pieces together into self-reliant local AI systems that perceive, act, and continually adapt on everyday computers, phones, and offline robots without depending on cloud infrastructure.